MID

Course: Data Structures (CS101)

Set: A | Time: 2 Hours | Max Marks: 37 | Date: 14-03-2026

Total Questions: 11 | Answer all questions.

Name: ____________________________ **Roll No:** _________________________

Student Sign: ___________________ **Teacher Sign:** ___________________

SECTION A: MCQs

1. Which layer of the OSI model is responsible for logical addressing (IP addresses)? **[2 Marks]**

a) Physical Layer
b) Data Link Layer
c) Network Layer
d) Transport Layer

2. Which of the following protocols is considered connection-oriented? **[2 Marks]**

a) UDP
b) HTTP
c) TCP
d) ICMP

3. What is the primary function of a modem in a computer network? **[2 Marks]**

a) To route data packets between networks.
b) To modulate and demodulate signals for transmission over analog lines.
c) To filter network traffic and improve security.
d) To connect multiple devices within a local area network.

4. A MAC address uniquely identifies a device on a network segment. What is its typical length? **[2 Marks]**

a) 16 bits
b) 32 bits
c) 48 bits
d) 128 bits

5. Which of the following transmission media is least susceptible to electromagnetic interference (EMI)? **[3 Marks]**

a) Coaxial Cable
b) Unshielded Twisted Pair (UTP)
c) Fiber Optic Cable
d) Shielded Twisted Pair (STP)

6. In the context of computer networks, what does LAN stand for? **[2 Marks]**

a) Local Area Network
b) Large Access Node
c) Link Area Node
d) Low-latency Access Network

SECTION A: SHORT ANSWERS

7. Define the primary purpose of the Transport Layer in the OSI model. **[3 Marks]**

8. Differentiate between a hub and a switch based on how they handle network traffic. **[5 Marks]**

9. Explain the fundamental concept of 'packet switching' in computer networks. **[3 Marks]**

10. Briefly describe two common network topologies and their main characteristics. **[6 Marks]**

11. Discuss the role of DNS (Domain Name System) in network communication. **[7 Marks]**

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